

Instructor  
Prof. B. Stellato

Als  
I. Wang, V. Ranjan, S. Clarke, J. Ge

**ORF307: Optimization**  
**Spring 2023 2<sup>nd</sup> Midterm Exam**  
**April 18, 11:00am – 12:20pm**

**Exam Rules**

1. Please write out and sign the following pledge on the solution booklet: “I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam.”
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. You should **always explain your answers** in the solution booklet.
6. There are 3 problems worth a total of 100 points.
7. At the end of the exam, please turn in both your solution booklet and the exam questions.

Name: \_\_\_\_\_

1. *Problem 1 (35 points)*. An  $\ell_\infty$  constrained problem.

Consider the following optimization problem:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && \|Ax - b\|_\infty \leq 1, \end{aligned} \tag{1}$$

for  $c \in \mathbf{R}^n$ ,  $A \in \mathbf{R}^{m \times n}$ ,  $b \in \mathbf{R}^m$ .

- (a) (10 points) Write problem (1) as an LP.
- (b) (15 points) Take the dual of the LP you derived in part (a). Please simplify your final answer to be in terms of  $c, A, b$ , and any new dual variables you create.

Now, let us plug in the following values to problem (1):

$$c = (-1), \quad A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad b = (3, 4).$$

For these values, the dual problem is equivalent to

$$\begin{aligned} & \text{maximize} && -d^T y \\ & \text{subject to} && Gy = h \\ & && y \geq 0, \end{aligned} \tag{2}$$

where

$$d = (3, 4, -3, -4, 1), \quad G = \begin{bmatrix} 1 & 2 & -1 & -2 & 0 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix}, \quad h = (1, 0).$$

For these values, the optimal solution to (2) is given by  $y^* = (0, 0.5, 0, 0, 0.5)$ .

- (c) (10 points) For the given problem data, what is the optimal basis  $B$ ? What is the optimal basis matrix  $G_B$ ? What is  $y_B^*$ ?

2. *Problem 2 (35 points)*. Consider the following optimization problem:

$$\begin{aligned} & \text{minimize} && -x_1 - 2x_2 \\ & \text{subject to} && 2x_1 + 3x_2 \leq 3 \\ & && -2x_1 + 3x_2 \leq 6 \\ & && x_1 \geq 0 \\ & && x_2 \geq 0. \end{aligned} \tag{1}$$

(a) (5 points) Write problem (1) as an LP of the following form:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \leq b. \end{aligned} \tag{2}$$

(b) (10 points) Write down the dual problem of the LP you derived in part (a). Expand your answer with the problem data instead of just writing the general dual formulation.

(c) (15 points) Show that feasible  $x = (0, 1)$  is optimal.

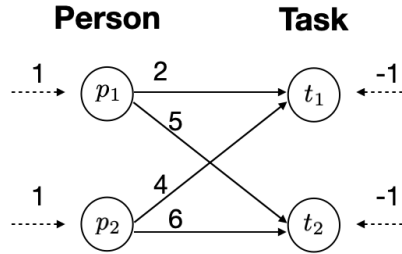
**Hint:** Use complementary slackness.

(d) (5 points) Let us modify  $b$  in (2) by adding  $u = (1, 0, 0, 0)$  to it, i.e., we replace  $b$  with  $b + u$ . Assuming that the optimal basis remains unchanged, what would be the new optimal value, denoted by  $p^*(u)$ , of (2) with the modified  $b$ ?

3. *Problem 3 (30 points).* A variant on the task assignment problem.

In the task assignment problem, we have  $n$  people and  $n$  tasks. Each person must be assigned to a task, and each task must be assigned to only one person. Additionally, for every assignment  $j$  of a specific person to a task, we incur cost  $c_j$ . The goal of this problem is to match people with tasks while minimizing the sum of costs.

- (a) (15 points) We can formulate this problem as a minimum cost network flow problem by setting up a network with  $n$  nodes representing the people, and  $n$  other nodes representing the tasks. Every arc  $i$  connects a person to a task and has cost  $c_i$ . Each person node has a *supply* of 1, and each task node has a *demand* of  $-1$ . The following image shows a simple example with  $n = 2$ . The people are labeled  $p_1, p_2$  and the tasks are labeled  $t_1, t_2$ . The supply and demands are written on the dashed arrows, and the costs are written *above* the solid arcs.



Formulate this specific example as a minimum cost network flow problem. That is, give  $c, A, b$  so that the optimization problem can be written as

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax = b \\ & && 0 \leq x \leq \mathbf{1}. \end{aligned} \tag{1}$$

- (b) (10 points) Now suppose we add a new task  $t_3$  but we maintain that each person can only perform one task, which means all tasks cannot be assigned. The cost associated with the arc connecting  $p_1$  to  $t_3$  is 7 and of the arc connecting  $p_2$  to  $t_3$  is 6. If a task  $t_j$  remains unassigned, then we pay a penalty cost  $d_j$ . Let  $d_1 = 10$ ,  $d_2 = 11$ , and  $d_3 = 12$ .

Our goal is to now minimize the sum of assignment costs plus any incurred penalties. How would you modify the network in part (a) to answer this new question? You may draw a new network for your answer instead of defining new matrices.

**Hint:** We need a way to encode an unassigned task. One approach is to introduce an imaginary person  $p_3$  and connect it by specifying arc costs and capacities.

- (c) (5 points) What if you have another new task  $t_4$ ? How would the network change? A word description of how you would change the network is sufficient.